

Decent Coaching Classes

VIII SB

Mind Your Math - VII

Marks: 40

Time: 2 Hrs.

Instructions: All questions are compulsory (internal choice as indicated). Figures to the right indicate full marks. Draw neat figures wherever necessary.

Q.1 (A) Choose the correct alternative.

[5]

- (1) The point of concurrence of the altitudes of a triangle is called the _____.
(A) centroid (B) orthocentre (C) incentre (D) circumcentre
- (2) One factor of $x^2 + 5x + 6$ is $(x + 2)$. The other factor is _____.
(A) $x - 3$ (B) $x + 4$ (C) $x + 3$ (D) $x - 2$
- (3) $a^3 - b^3 =$ _____.
(A) $(a + b)(a^2 - ab + b^2)$ (B) $(a - b)(a^2 + ab + b^2)$
(C) $(a - b)(a^2 - ab + b^2)$ (D) $(a + b)(a^2 + ab + b^2)$
- (4) If x varies directly as y and $x = 3$ when $y = 12$, the constant of variation ($x = ky$) is _____.
(A) $1/4$ (B) 4 (C) 36 (D) 9
- (5) Which of the following pairs is in inverse variation?
(A) pens and their total cost (B) weight of rice and its cost
(C) distance and petrol used (D) workers and time to finish a job

Q.1 (B) Solve the following.

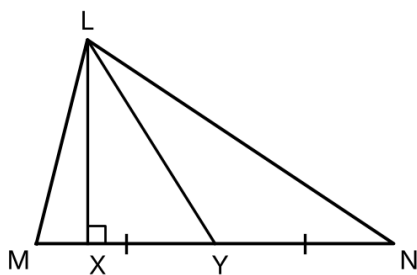
[3]

- (1) Name the point of concurrence of the medians of a triangle.
- (2) Write the factors of $a^3 + b^3$.
- (3) If the speed of a vehicle is doubled, what happens to the time taken to cover the same distance?

Q.2 Solve any four of the following.

[8]

- (a) In the figure given below, seg LX and seg LY are drawn from vertex L of $\triangle LMN$ to side MN. Write which of them is an altitude and which is a median.



- (b) Write the position of the orthocentre of (i) a right angled triangle (ii) an obtuse angled triangle.
- (c) Factorise: $x^2 + 8x + 12$
- (d) Factorise: $y^2 - 7y + 12$
- (e) a varies inversely as b . When $a = 6$, $b = 20$. Find a when $b = 10$.
- (f) Write the following statements using the symbol of variation: (i) The pressure (p) of a liquid varies directly as its depth (d). (ii) The wavelength (λ) and the frequency (f) of sound are in inverse variation.

Q.3 Solve any four of the following.

[12]

- (a) Write the definitions of altitude and median of a triangle. Name the point of concurrence of the altitudes and of the medians.
- (b) Factorise: $2x^2 + 7x + 3$
- (c) Factorise: $27p^3 + 8q^3$
- (d) Factorise: $8m^3 - 27n^3$
- (e) The cost of sugar is directly proportional to its weight. If 5 kg of sugar costs ₹200, find the cost of 12 kg of sugar.
- (f) A car running at a speed of 60 km/hr takes 4 hours to complete a journey. How much time will it take to complete the same journey at a speed of 80 km/hr?

Q.4 Solve any three of the following.

[12]

- (a) Draw an acute angled $\triangle PQR$. Draw all of its altitudes. Name the point of concurrence as O.
- (b) Draw an obtuse angled $\triangle STV$. Draw all of its medians and show the centroid as G.
- (c) Simplify: $(x^2 - 5x + 6) / (x^2 - 4)$
- (d) Factorise: (i) $3x^2 - 12x - 15$ (ii) $27x^3 - 1$
- (e) 20 workers can complete a piece of work in 12 days. Identify the type of variation and find how many workers are required to complete the same work in 8 days.